Fitting an Authenticated Emergency Report into One BLE Advertising Frame: Semantic Compression for Infrastructure-Free Disaster Mesh Networks

Gauri Dhingra
Dept. of CSE (AI&ML)
KIET Deemed to be University
Ghaziabad, India
gaurdhingra0@gmail.com

Ity Gaur
Dept. of CSE (AI&ML)
KIET Deemed to be University
Ghaziabad, India
itygaur@gmail.com

Mahima Bhat
Dept. of CSE (AI&ML)
KIET Deemed to be University
Ghaziabad, India
mahimabhat2@gmail.com

Nischhal Garg
Dept. of CSE (AI&ML)
KIET Deemed to be University
Ghaziabad, India
nischhalgarg61@gmail.com

Preeti Singh
Dept. of CSE (AI&ML)
KIET Deemed to be University
Ghaziabad, India
preetisinghthakur07@gmail.com

Abstract

When cellular infrastructure fails, smartphone mesh networks can still carry distress reports hop by hop over Bluetooth Low Energy. On the handsets most common in disaster-prone regions the only universally supported channel is legacy BLE advertising, which carries 31 bytes per frame and about 27 after the mandatory AD structures. An authenticated emergency report does not fit: in our system it occupies 124 bytes before a single character of free text, because an Ed25519 public key and signature account for 96 of them. We present a semantic compression path that fits a complete authenticated report into one frame. Free text is encoded on the handset into a 3-byte envelope carrying category, casualty count, mobility, hazard flags, injury severity and the encoder's own confidence; the message is then packed into 25 bytes alongside a coarse grid location and a 4-byte authentication tag. Reaching that budget required reducing cryptographic overhead as well as text, and we state each trade explicitly. Below a confidence threshold nothing is compressed and the original text is transmitted. We extend a discrete-event mesh simulator to model frame-level fragmentation, size-dependent airtime and receiver-side collisions, none of which it previously represented. Across 60 trials per configuration, compression raises the fraction of reports reaching a gateway from 53.3% to 60.0% at 200 nodes and cuts channel occupancy by 8.5–15.7%. We also report two negative results: a model trained on synthetic data reaches 99.7% on its own held-out split and loses to a keyword baseline on unseen text, though a cascade between them reaches 67.7% against 59.5% and 53.4% alone; and a sample of 520 tweets from a standard crisis corpus proved 51% commentary and 18% news reporting, a mismatch of population rather than of annotation quality.

Index Terms— disaster communication, BLE mesh, semantic compression, on-device NLP, emergency triage, code-mixed text

I. Introduction

Earthquakes, floods and building collapses remove the infrastructure that emergency communication depends on. Base stations lose power, backhaul is cut, and the handsets that survive have no network to reach. Smartphone mesh networking offers a way through: phones relay for one another over short-range radio until a message reaches a device that still has connectivity.

The constraint that shapes such a system is not bandwidth in the usual sense. It is the advertising frame. BLE connectionless advertising is the only mode that works without pairing, without a connection, and on every Android handset in the field, including the five-year-old devices that dominate the regions where these systems would be deployed. A legacy advertising PDU carries 31 bytes; after the mandatory Flags structure and a 16-bit Service Data header, roughly 27 remain.

Our heartbeat message was engineered to 25 bytes and fits. The SOS message does not. Measured on our implementation, an SOS occupies 124 bytes carrying no text at all, rising to 162–187 bytes with a short description — six or seven frames. The reason is not the text. An Ed25519 public key (32 B) and signature (64 B) account for 96 of those 124 bytes, 77% of the message, before anything is said about the emergency.

This matters because fragmentation is not free. A message split across n frames arrives only if every frame arrives: at 3% per-frame loss, a six-frame message fails 16.7% of the time where a single-frame message fails 3%. It also occupies the channel longer, and in a dense mesh where every node relays, channel occupancy limits how many reports get through.

Fig. 2 shows the path a report takes. The question this paper asks is therefore narrower than whether NLP can help disaster response. It is: *can a complete, authenticated emergency report be made to fit in one legacy BLE advertising frame, and what does that buy at the network layer?*

A. Contributions

A semantic envelope sized against a derived budget. We compress free-form emergency text into 3 bytes carrying six fields and fit the complete authenticated message into 25. The budget follows from the advertising PDU structure rather than being asserted, and reaching it required reducing cryptographic overhead as well as text — a trade prior work has not had to confront because it does not target a frame boundary.

A size-aware mesh simulator, and network-level results. We add frame-level fragmentation, size-dependent airtime and receiver-side collisions to a discrete-event simulator that previously charged every transmission a fixed 5 ms and dropped whole messages with one coin flip, so payload length could not affect any outcome. We then measure what compression does to delivery, not only to classification accuracy.

Negative results reported in full. A model trained on synthetic data outperforms a keyword baseline by nearly ten times on its own held-out split and loses to it on hand-written text. Stratifying 10,000 messages by distance from the lexicon shows the two fail in opposite places, and a cascade between them beats either. We also document that a standard crisis corpus does not contain the kind of message this system carries.

II. Related Work

A. Classification of crisis text

Automatic categorisation of disaster-related messages is mature. HumAID [1] and the consolidated CrisisBench benchmark [2] provide standard splits and transformer baselines for humanitarian category, informativeness and priority labelling. Subsequent work has largely refined accuracy on these benchmarks rather than altering what is transmitted. We treat crisis-text classification as available infrastructure rather than as a contribution, and adopt it only as the front end of an encoder whose output is a transmission format rather than a label.

B. On-device intelligence in offline mesh networks

The closest work is ResQMesh [3], which extends an open-source BLE mesh messenger with on-device machine learning. Its priority module combines roughly ninety keyword patterns with a quantised INT8 Conv1D TensorFlow Lite classifier, assigns nine categories and four urgency tiers, and reports a composite macro-F1 of 0.920 on 405 held-out messages at 0.19–1.97 ms. Classified messages enter a priority queue so critical traffic preempts other traffic at the radio layer.

ResQMesh establishes that useful semantic processing is feasible on commodity Android hardware without connectivity, and we build on that premise. **Our work differs in what the classifier is for.** ResQMesh uses the inferred category to *order* transmissions; the message itself is unchanged, and no payload size, MTU, advertising budget or fragmentation behaviour is reported. A message too large for one frame remains too large after classification, and the reported latency improvement is measured at the queue rather than over the air. We use the inferred semantics to *constrain the payload* and evaluate the consequence at the network layer.

C. Semantic communication and extreme compression

Semantic communication reframes transmission around conveying meaning rather than reconstructing symbols. Recent work examines tiny language models under a joint latency–accuracy–size constraint [4], and semantic text compression evaluated by preserved downstream classification rather than reconstruction fidelity [5]. This literature assumes a channel with meaningful capacity; to our knowledge none targets a single-digit-byte residual budget inside a 31-byte advertising PDU. In the grey literature, RelayZero [6] proposes offline edge-AI semantic compression for disaster mesh using a locally hosted language model. We note it as concurrent evidence of interest and distinguish our contribution: it reports no accuracy, latency or reliability measurement, its mesh layer is a local TCP socket rather than a radio, its stated 25-byte payload is inconsistent with its own published example string, and its encoder is not deployable on the handsets we target.

D. Authenticated offline emergency broadcast

The Offline Emergency Peer-to-Peer Broadcast Protocol (OEPB) [7], an experimental Internet-Draft, specifies authenticated emergency alerts over short-range radios: a 256-byte CBOR datagram with a 40-byte fixed header and an optional 64-byte Ed25519 signature. Its existence confirms independently that authenticated infrastructure-free emergency broadcast is worth specifying. **Its answer is the one we set out to avoid.** On BLE 4.x, OEPB states that the 31-byte advertisement payload requires L2 fragmentation and reassembly; its signed overhead alone is 104 bytes, and free text is carried as plain UTF-8 under a byte cap with no semantic compression. We ask whether that fragmentation can be avoided rather than accommodated.

E. Short authentication tags on constrained links

Our 4-byte tag has prior art. Cumulative Message Authentication Codes [8] address the same tension: a full MAC does not fit, so a 128-bit tag is split into segments spread across successive messages, with the receiver accumulating strength over time. We adopt the truncated tag without the accumulation, and note the omission as the clearest available improvement: our store-and-forward layer already re-broadcasts each SOS every ten seconds for up to six hours, so the repeated transmissions such a scheme requires are present and currently carry an identical tag each time.

F. Inference from absence of signal

Post-disaster victim localisation is typically a sensing problem, addressed with millimetre-wave radar, UAV-mounted sensors, or passive detection of emissions from victims' handsets [10]. These detect a victim's presence. The silence-inference component of the system this work extends inverts the premise: participants carry a cooperating application emitting periodic heartbeats, and a disaster is inferred from the *correlated cessation* of those heartbeats within a spatial cluster, discriminated against battery depletion, deliberate shutdown, a single participant walking out of range, and whole-network partition.

G. Code-mixed emergency text

Our deployment context produces Hindi–English code-mixed input. Pretrained resources exist, including L3Cube-HingCorpus and HingBERT [11], but have not to our knowledge been applied to emergency triage under a transmission-size constraint.

III. Background: The Byte Budget

A. Why legacy advertising

Bluetooth 5 extended advertising carries up to 254 bytes per packet and 1,650 via chaining, but requires a Bluetooth 5.0 chipset and OEM driver support. Budget and older Android handsets — precisely those most common in disaster-prone regions — support only legacy advertising. Wi-Fi Direct and Nearby Connections offer higher throughput but are likewise not universally available and cost substantially more energy. A system whose emergency path depends on any of these degrades to nothing on the devices most likely to be present. We therefore treat legacy advertising as the floor every message must clear, and faster transports as an optimisation when available.

B. Deriving the usable payload

TABLE I
USABLE PAYLOAD OF A LEGACY ADVERTISING PDU

Element	Bytes
Advertising PDU payload	31
Flags AD structure (`0x02 0x01 0x06`)	−3
Service Data AD header (length, type, 16-bit UUID)	−4
Usable for application data	**27**

C. What the existing messages cost

TABLE II
MESSAGE SIZES BEFORE COMPRESSION

Message	Bytes	Frames at 27 B
Heartbeat	25	1
SOS, no text	124	5
SOS, 13 B text	137	6
SOS, 37 B text	161	6
SOS, 200 B text	324	12

The heartbeat was designed to the budget and fits. The SOS does not, and the reason is cryptography rather than content: a 13-byte header, a 15-byte body, a 32-byte public key and a 64-byte signature give a 124-byte floor. Fig. 1 shows this against the budget. **Compressing text alone moves a typical report from six frames to five.** It cannot reach one.

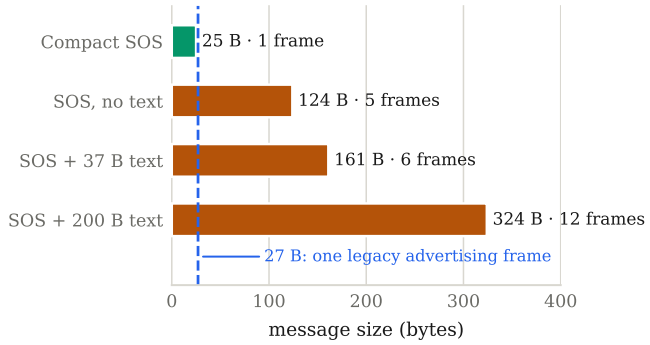


Figure 1. Message size against the frame budget. Everything above the dashed line fragments; 96 of the 124-byte floor is the public key and signature, so compressing text alone cannot cross it.

D. Why frame count matters

Compound loss. A fragmented message arrives only if every fragment arrives. With independent per-frame loss p and n frames, delivery probability is $(1-p)^n$. At $p = 0.03$, one frame fails 3% of the time and six frames fail 16.7% — a 5.6× increase in failure for the same report.

Channel occupancy. On the 1M PHY each bit takes 1 µs, and each frame carries 10 bytes of PDU overhead plus an inter-frame space. A 25-byte message occupies about 0.43 ms; a 364-byte message across 14 frames occupies about 6.1 ms. In a flooding mesh this difference multiplies across the network and raises the chance that two frames overlap at a receiver.

IV. System Design

A. The semantic envelope

Free text is encoded into 24 bits: category (4), casualty count (3, where 7 means seven or more), mobility (2), hazard flags (6), severity (2), the encoder's own confidence (3), and 4 reserved bits. Confidence travels *with the data* deliberately: a rescuer must see how much of a report was inferred rather than stated, and the relay layer uses the same value to decide whether to compress at all.

B. Reaching 27 bytes

Text compression alone is insufficient. Three further reductions were necessary, each a trade we state rather than hide. The public key is resolved from `node_id` at the gateway rather than transmitted (−32 B), so only enrolled nodes can send compact messages. The 64-byte Ed25519 signature becomes a 4-byte truncated HMAC (−60 B); this is *not* a digital signature, it authenticates only to a gateway holding the shared key, and a 32-bit tag admits a one-in-four-billion forgery per attempt. Float latitude and longitude become the ∼100 m grid cell the heartbeat already uses (−4 B), giving rescuers a cell rather than a pin, with ∼11 m quantisation error observed.

The tag excludes TTL. TTL decrements at every hop, so a tag computed over the live header would verify only at the originating node. Zeroing it means the tag authenticates the

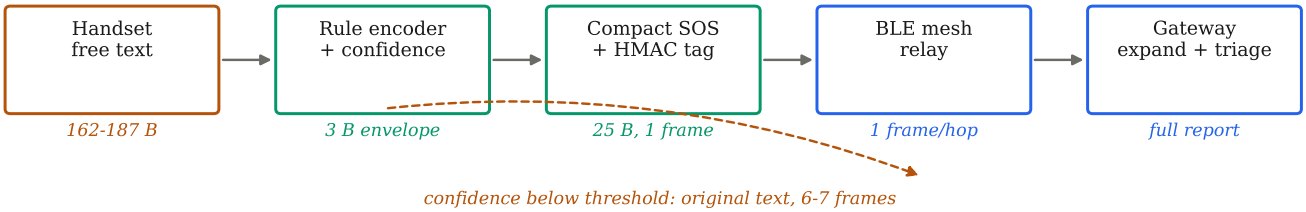


Figure 2. The compression path. A report is encoded on the handset into three bytes and sent as a 25-byte authenticated frame; the gateway expands it for triage. When the encoder's confidence falls below threshold nothing is inferred and the original text is sent instead, at six or seven frames.

TABLE III
THE COMPACT SOS MESSAGE

Element	Bytes
Header (version, type, node id, seq, TTL, timestamp)	13
Grid x, grid y	4
Battery	1
Semantic envelope	3
Authentication tag	4
Total	**25**

report rather than the path it travelled. The grid is relative to a deployment origin distributed at enrolment: absolute coordinates do not fit, since longitude 77.209 at 0.001° resolution is 77,209, past the 32,767 ceiling of a signed 16-bit field. Both SOS layouts share the message type; the version byte distinguishes them.

C. *Threat model*

Four bytes of authentication invites an immediate objection, so we state precisely what the tag defends, against whom, and what it knowingly does not. The asset is the integrity of the rescuer's view: a forged SOS moves rescuers to an empty field. Confidentiality is deliberately absent — an encrypted cry for help trades availability in exactly the wrong place — so anyone in radio range learns that grid cell (x, y) reported trouble; we state this privacy cost rather than hide it.

The adversary transmits and receives freely but holds no key. Blind forgery succeeds per attempt with probability 2^{-32}; under the airtime model of Section III a 25-byte frame occupies 430 µs, bounding one channel at ∼2,300 attempts/s, so even an attacker saturating the channel — a visible denial of service in itself — expects a first forgery after $2^{32}/2300 \approx 21$ days of continuous transmission, and the gateway can rate-limit verification failures per node id. Replay is bounded because the tag covers `node_id`, `seq` and `timestamp` (only TTL is zeroed), and both relays and gateway deduplicate on (`node_id`, `seq`).

Two honest weaknesses remain. The prototype uses a single group key, so one captured device forges any identity; the deployment design derives $K_{node} = \mathrm{KDF}(K_{master}, node_id)$ at enrolment, at zero byte cost, after which a compromise forges only itself. And relays, holding no key, amplify injected frames with fresh sequence numbers; suppression (k=2) and TTL bound the amplification, a per-node token bucket at relays is designed but unimplemented, and an attacker with a transmitter could in any case jam the band outright. Tag comparison is constant-time at both endpoints. The clearest upgrade is CuMAC-style accumulation (Section II): rotate four segments of a 16-byte tag across the store-and-forward re-broadcasts the system already performs, reaching 128-bit strength over repeats at zero bytes per frame.

D. *The encoder*

The deployed encoder is rule-based: lexicon matching over category, casualty, mobility, hazard and severity terms, with numeric extraction that binds a count to a nearby person-word so "4 people trapped" is read and "4 floors" is not. Lexicons cover English and romanised Hindi together, since code-mixed input is the norm rather than an edge case. Negation is checked on both sides of a match, because English negates before the term ("no injuries") and romanised Hindi after it ("*ghayal nahi*"). Confidence combines how many of the five fields were recovered with how much of the message the lexicons failed to account for: **a message carrying much the encoder could not read lowers confidence rather than raising it**.

E. *Fallback*

Below a confidence threshold of 0.45 the sender transmits the original full-text SOS. Nothing is inferred that the encoder did not understand.

V. EVALUATION

A. *Making the simulator size-aware*

The simulator this work extends could not have measured compression. Every transmission was charged a fixed 5 ms regardless of length, whole messages were dropped with a sin-

gle coin flip, and there was no collision model. We verified the consequence: the same 200-node scenario run with an 11-byte and a 240-byte message produced byte-identical results — 0 delivered, 2,474 transmissions, 5,588 dedupe drops in both. We added airtime derived from length, fragmentation above 27 bytes with independent per-frame loss, and collisions between frames overlapping at one receiver, each switchable so prior behaviour remains reproducible.

A note on a prior published figure. Our earlier evaluation reported 0 SOS delivered at 200 nodes. That is a single-seed artefact: seed 42 places the source 17 hops from the nearest gateway against a TTL of 8, while the median across seeds is 6. All results below average over seeds.

B. *Encoder fidelity, stratified*

One number over a generated set would report how the generator happened to phrase things. We therefore split 10,000 messages by how far the wording sits from the encoder's lexicon, with labels true by construction.

TABLE IV
RULE ENCODER BY STRATUM (2,000 MESSAGES EACH)

Stratum	Category	Count	All	Fallback
known_vocab	79.8%	100.0%	58.6%	16.2%
synonyms (OOV)	8.3%	60.3%	0.1%	97.0%
spelling	69.8%	100.0%	50.9%	25.2%
code_mixed	79.9%	100.0%	60.4%	17.4%
unintelligible	—	—	—	100.0%

The out-of-vocabulary cliff is total, and it is safe. On synonyms the encoder recovers the category 8.3% of the time, and refuses to compress 97.0% of them. The failure mode is slower transmission, not a wrong report. Code-mixed input is not the weak point: at 60.4% it edges out pure known-vocabulary text. Refusal on unintelligible input is perfect across 2,000 messages.

C. *A learned encoder*

The real corpora were unreachable from our environment, so we generated 6,000 messages from templates whose field values are known by construction — ground truth rather than rule output — and trained a character n-gram TF-IDF model with one logistic-regression head per field. It reaches 99.7% exact-envelope accuracy on held-out synthetic data against the rules' 10.3%, and 28.0% against the rules' 76.0% on hand-written messages, at 808 KB and eight times the latency. The model learned the generator, not the task.

D. *The two encoders fail in opposite places*

Scored per stratum, the model does not simply lose: on out-of-vocabulary wording it recovers nearly five times what the lexicon does, because character n-grams reach words no keyword list contains. The rule encoder already emits a confidence deciding whether to compress; routing on it requires nothing further

TABLE V
CATEGORY ACCURACY: RULES, MODEL, AND A CASCADE BETWEEN THEM

Stratum	Rules	Neural	Cascade
known_vocab	79.8%	62.3%	**81.2%**
synonyms (OOV)	8.3%	**39.1%**	38.8%
spelling	69.8%	61.7%	**74.8%**
code_mixed	**79.9%**	50.5%	75.8%
Overall	59.5%	53.4%	**67.7%**

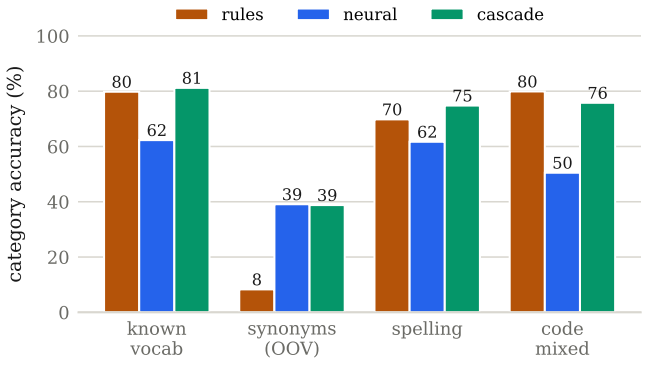


Figure 3. The two encoders fail in opposite places. The lexicon wins where it has the words; the model wins on wording it has never seen. Routing between them on the confidence the rule encoder already emits beats either alone.

to be trained. The unintelligible stratum is excluded from the overall figure, since its gold label is OTHER and both encoders answer OTHER. Unintelligible input is still refused 100% of the time.

E. Why not general-purpose compression

TABLE VI
COMPRESSING THE TEXT INSTEAD OF ENCODING IT

Method	Mean B	Max	Fits 27 B?
Raw UTF-8	48.2	65	no
zlib (level 9)	53.8	66	no
gzip (level 9)	65.8	78	no
bzip2	77.0	92	no
LZMA	105.3	124	no
zlib + shared dictionary	11.2	51	sometimes
Semantic envelope	**3.0**	**3**	**always**

General-purpose compressors *enlarge* messages this short: their headers exceed what they save. The serious competitor is zlib with a shared dictionary, which averages 11.2 bytes and is lossless. We distinguish it on three grounds. Its output is variable — up to 51 bytes here — and a frame budget is a bound, not an average. It yields bytes rather than fields, so the dashboard must still parse text to rank by urgency. And it requires the dictionary to stay synchronised: a mismatch destroys the message rather than degrading it. It is, however, strictly better than the raw text our fallback path sends, and compressing that path would shrink the 23% of messages taking it.

F. Network effect

Three conditions, 60 trials per cell pairing seed and message round-robin, TTL 8, 3% per-frame loss: **A** full text, **B** always compressed, **C** compressed when confident.

TABLE VII
SOS DELIVERED TO A GATEWAY

Nodes	A	B	C	Airtime saved (C)
50	80.0%	88.3%	86.7%	15.7%
100	76.7%	85.0%	80.0%	12.0%
200	53.3%	60.0%	60.0%	8.5%

The gain is largest at 200 nodes, where the mesh is most loaded. **C is the shipping configuration and the number to quote**; B assumes the encoder is never unsure, which is false 23% of the time. A separate sweep separates TTL-limited from size-limited failure: where TTL is too low for the topology, both short and long messages fail and compression cannot help.

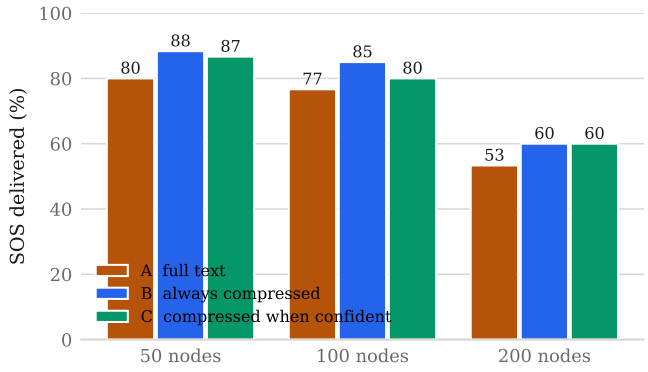


Figure 4. SOS reports reaching a gateway, by condition and network size. The margin widens as the mesh loads: compression buys the most where contention is highest.

VI. DISCUSSION

A. What the negative result means

Template-generated text teaches a model the template. On its own split that model looks close to perfect; on text written by a person it collapses below a keyword matcher. Crisis-NLP systems are frequently evaluated on held-out splits of the corpus they were trained on, and where that corpus is narrow the reported accuracy may say more about the corpus than the system. Our result is an extreme case because our corpus is synthetic, but the direction of the error is the same one a single-event benchmark would produce.

B. Available corpora collect the wrong half of the conversation

We attempted to replace generated text with real data and did not succeed. HumAID [1] supplies 76,484 annotated tweets from 19 disasters. We drew 520, weighted toward the classes most likely to contain an emergency, and classified what we had: 51% commentary, 18% news reporting, 17% retweets, 1% donation appeals. The 13% matching first-person patterns were largely expressions of sympathy rather than requests for help.

This is not a defect in HumAID. It is a mismatch of population. Crisis corpora are collected from public social media, where the people writing are overwhelmingly *observers*. A mesh system of this kind carries messages from *participants*. The two differ in length, register, person, and the presence of hashtags and audience, and the labels of the first describe information type rather than emergency type. The Figure Eight *Disaster Response Messages* corpus [9] carries a `direct` genre of messages and SMS sent during the 2010 Haiti and Chile earthquakes, a closer population, though translated and smaller.

C. The safety case

A compression error here is not a metric: reading “two children trapped” as “one adult trapped” sends the wrong team. Three bits of the envelope carry how certain the encoder was, so a rescuer’s display can mark an inferred report as inferred. A

message where the lexicons account for little of the text scores *lower*, not higher — the encoder treats its own incomprehension as evidence against compressing. Below threshold the original text is transmitted at the cost of more frames, a deliberate exchange of reliability for fidelity in the direction that fails safe. A regression test asserts that casualty counts are never reported lower than ground truth. We make no claim that these eliminate the risk.

VII. Limitations

No real evaluation corpus. Fidelity is measured on generated text and 25 hand-written messages. Our attempt to use a real corpus found a population mismatch (§ VI-B). **No hardware validation.** Everything is simulated; the encoder has not run on a handset and no energy measurement exists. **A simplified radio model.** No multipath, capture effect, or hardware-specific advertising behaviour. **Weaker authentication.** The 4-byte tag is not a digital signature and the shared key in our prototype is a placeholder that must not ship. **Coarser location.** A $\sim$100 m cell rather than a GPS pin. **Enrolment required.** An unenrolled handset cannot send a compact SOS. **Single implementation.** Results have not been reproduced against an independent BLE stack.

VIII. Conclusion

An authenticated emergency report can be made to fit in one legacy BLE advertising frame, but not by compressing text alone — cryptographic overhead dominates the message, and reaching 27 bytes required reducing it too. Doing so raises the share of reports reaching a gateway from 53.3% to 60.0% at 200 nodes and cuts channel occupancy by 8.5–15.7%, with a confidence-gated fallback that transmits the original text on the 23% of messages the encoder does not understand.

The encoder that achieves this is a keyword matcher. A learned model trained on synthetic data scored 99.7% on its own held-out split and 28.0% on hand-written text. But the two fail in opposite places, and a cascade routed on a confidence signal that already existed reaches 67.7% against 59.5% and 53.4% alone. We report both negative results in full because quoting the synthetic column would have been a false finding, and because together they say what the next step needs: not a better architecture, but emergency text written by people in emergencies, with the envelope fields annotated. We looked for it and found that the standard corpora collect the other half of the conversation.

Artefact Availability

All code, measured results and reproduction commands are in the project repository.

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